October 21, 1958

FILE MEMO

SUBJECT: Stretch Performance

As part of a recent study of floating point addition seven problems were traced on the 704 and a frequency count of the operations executed was made.

I have grouped the various 704 operations into blocks which are roughly equivalent to similar classes of Stretch instructions. This grouping is as follows:

A	All divide	DVH, DVP, FDH, FDP
B	All multiplies	FMP, MPR, MPY, UFM
C	Floating adds	FAD, FSB, UFA, UFS
D	Loads and stores	CLA, LDQ, STO, STQ
E	All Index ops	
F	Test Operations	CHS, LBT, MSE, NOP, PBT, PSE, SSM, SSP, and all non-index transfers.
G	All other ops	

Table 1 gives the results for the total of all problems traced (equal weight by execution). Table 2 gives the results for the SNG code from Los Alamos. This code is of particular interest since it will probably be one of the test problems.

Using the execution times shown in the tables, I find that the average 704 instruction execution time is 44 usec. for all problems and 58 usec for the SNG code. This increase is due almost exclusively to a higher percentage of floating multiply and add operations.

If we assume that Stretch has 100% arithmetic efficiency and all index operations and tests are overlapped with arithmetic we find an an average instruction execution time of 0.55 usec for all problems and 0.72 usec for SNG. In both cases this gives an 80X computer. We are fairly sure that the Stretch instructions will allow us to write programs with about 20% fewer instructions. (about 10% of executed 704 instructions are LDQ, STQ, LRS, LLS; the universal accumulator in Stretch will eliminate almost all of these) Applying this factor we then have under ideal conditions about a 100X computer.

Allowance at this stage must be made for the fact that the arithmetic unit is only about 60% efficient not 100% as was assumed. This will then give us a 60X computer which checks well with Dr. Kolsky's reports on the use of the timing simulator. (The 60% efficiency figure is taken from his second report).

Another important figure available from these statistics is the percentage of instructions tagged. The following table shows the percentage of each category which were tagged.

	All Problems	SNG
All ops.	44%	50%
Index ops	98%	100%
Non-index ops	28%	38%

D. W. Sweeney
Product Planning Coordinator
Project 7000

DWS/jcv
Enclosure Table I and II

cc: Mr. S. W. Dunwell
Mr. R. E. Merwin
Mr. E. Bloch
Mr. D. W. Pendery
Mr. H. G. Jones
Dr. H. G. Kolsky

TABLE I - All Problems

(Time in Microseconds)

OPER		%Exec	704 Time Exec	704 Time	%Time	Stretch Time Exec.	Stretch Time	% Time
F DIV	A	2.0	216	4.32	9.9	7.5	.150	27.5
F MPY	B	5.6	204	11.42	26.2	2.0	.112	20.5
F ADD	C	9.5	84	7.98	18.3	1.0	.095	17.4
LD-ST	D	28.5	24	6.84	15.7	0.4	.114	20.9
INDEX	E	22.5	24	5.40	12.4	0.0	.000	0.0
TEST	F	13.2	24	3.17	7.3	0.0	.000	0.0
OTHER	G	18.7	24	4.49	10.2	0.4	.075	13.7
				43.62			.546	

TABLE II - SNG Code

F DIV	A	1.9	216	4.10	7.1	7.5	.142	19.7
F MPY	B	10.4	204	21.22	36.7	2.0	.208	28.8
F ADD	C	19.1	84	16.04	27.7	1.0	.191	26.5
LD-ST	D	29.8	24	7.15	12.4	0.4	.119	16.5
INDEX	E	19.6	24	4.70	8.1	0.0	.000	0.0
TEST	F	3.6	24	.86	1.5	0.0	.000	0.0
OTHER	G	15.6	24	3.74	6.5	0.4	.002	8.5
				57.81			.722	